Supplementary Material for VERA-ECG: Verifiable Evidential Reasoning and Anatomy-Grounded Multimodal Framework for Reliable 12-Lead ECG Interpretation

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1 Implementation details

Table 1: Training and evaluation settings.

Item	Setting
Framework	PyTorch 2.14, CUDA 13.0, Python 3.14
GPU	NVIDIA RTX 4060 Ti, 16 GB
A0-mV	AdamW, lr 10^{-3}, wd 10^{-2}, batch 128, 100 epochs, patience 15
A1-Pre	AdamW, lr 10^{-4}, wd 0.05, batch 32, 30 epochs, patience 8
Stage 2 / APEB	AdamW, lr 10^{-3}, wd 10^{-2}, batch 512, 100 epochs, patience 15
A8-OneWay	$\lambda_{\text{logic}} = 0.1$, $\delta = 0.1$, stop-gradient on diagnosis
C0/C2/C3 heads	AdamW, lr 10^{-3}, wd 10^{-4}, batch 512, 100 epochs
Qwen SFT	LoRA $r = 16$, $\alpha = 32$, dropout 0.05, lr 2×10^{-4}
Qwen batching	batch 1, gradient accumulation 8, max length 768, 1 epoch = 3,584 steps

2 CPSC2018 preprocessing

CPSC2018 waveforms were resampled from 500 Hz to 100 Hz with polyphase anti-aliasing using `scipy.signal.resample_poly(x, 1, 5, axis=1)`. No additional μV-to-mV factor was applied because the released arrays used in this study were already mV-scale (range approximately -4 to $+3$). Records shorter than 1,000 samples were zero-padded; longer records were center-cropped to 1,000 samples. PTB-XL arrays were used at 100 Hz. Per-lead z-score normalization was applied except for the mV-statistics branch, which receives the 36-dimensional global voltage descriptor before normalization.

3 A8-OneWay logic weights

The B2/B3 auxiliary validity and bucket heads were not trained; their contradiction and validity values are therefore not used as baselines. A7 (full APEB without logic loss) reaches 0.88% contradiction, and A8-OneWay reaches 0.68% (three-seed mean).

Table 2: A8-OneWay and bidirectional A8 weight sweep (seed 42).

Mode	λ_{logic}	AUC	AP	F1	Contradiction	Group F1
A8-OneWay	0.00	0.9211	0.8114	0.7365	0.88%	0.5353
A8-OneWay	0.05	0.9211	0.8114	0.7365	0.79%	0.5347
A8-OneWay	0.10	0.9211	0.8114	0.7365	0.79%	0.5347
A8-OneWay	0.20	0.9211	0.8114	0.7365	0.79%	0.5348
A8 (bidirectional)	0.05	0.9198	0.8074	0.7346	0.88%	0.5384
A8 (bidirectional)	0.10	0.9198	0.8072	0.7359	0.74%	0.5375

4 HE-DVR full-data ablations

All variants were retrained for one full epoch (28,672 training pairs, 3,584 steps) and evaluated on all 6,330 ECG-QA test questions.

Table 3: Full-data retrained HE-DVR ablations.

Variant	Legacy EM	Set EM	Set F1
Full (soft A/B/D + hard anchor)	**63.63**	**59.83**	**61.99**
No hard anchor	56.27	54.66	55.50
No image soft tokens (anchor retained)	62.76	58.74	61.12
Signal-only (no image, no anchor)	55.20	53.03	54.37

5 Cross-center details

C0 is trained only on PTB-XL training labels and applied to each CPSC fold without target labels or target statistics. C2/C3 use unlabeled CPSC fold features; they are transductive adaptation, not pure zero-shot transfer. Temperature $T = 2.507$ is fitted on the PTB-XL validation split. Few-shot experiments use only Fold 0 training labels and test Fold 0 records. The public CPSC folds provide filenames but no patient identifiers; only fold-level separation can be asserted.

C0 fold AUCs: 0.9080, 0.9092, 0.9091, 0.9071, 0.9185 (mean 0.9104 ± 0.0041). The same-fold Fold 0 comparison is Old A0 0.8603 vs. C0 0.9080 (+4.77 pp).

6 Uncertainty and confidence intervals

TMC clean test uncertainty predicts sample-level error with AUROC 0.738 (95% bootstrap CI [0.717, 0.760]); error at 10% coverage is 5.1%, and at full coverage 38.2%. The A8-OneWay contradiction rate at seed 42 is 0.79% (95% bootstrap CI [0.46%, 1.16%]). The counterfactual CSR is 7.12 (95% CI [4.20, 24.38]) in probability space and 8.70 (95% CI [6.01, 15.99]) in logit space. ECG-QA full Set EM is 59.83% (95% Wilson CI [58.62%, 61.03%]).

7 Data and code availability

All datasets are public and de-identified: PTB-XL, CPSC2018, and ECG-QA. The ECG-QA train/test split has zero patient-ID overlap (9,100 vs. 2,444 patients). Code, split definitions,

and evaluation scripts will be released in a public repository upon acceptance; an anonymized review link will be provided in Editorial Manager.